

SANJEEV RAJ

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PROFILE

Aspiring! Gen AI and Machine Learning professional with hands-on experience in building, deploying, and optimizing machine learning and deep learning applications. Skilled in Python, SQL, TensorFlow, FastAPI, Docker, and AWS EC2 with practical exposure to NLP, computer vision, and object detection projects. Experienced in deploying end-to-end ML solutions and working with real-world datasets. Seeking an opportunity to contribute to a data-driven organization as a Junior Data Analyst, Machine Learning Engineer, or Python Developer.

SKILL

Programming: Python, SQL

Ai & Machine Learning: Retrieval-Augmented Generation (RAG), Multi-Agent AI Systems, Random Forest, XGBoost

Frameworks & Libraries : TensorFlow, Keras, Scikit-learn, Langchain, LangGraph, FastApi, Flask, React.js, Next.js

Databases, Vector Stores & LLMs : PostgreSQL, pgvector, FAISS, Qdrant, Gemini API

Cloud & Deployment : Docker, AWS EC2, GCP, Git, Streamlit

PROJECTS

AI Multi-Agent Coding Assistant

- Developed an intelligent multi-agent coding assistant using LangGraph and LangChain for autonomous task orchestration.
- Built specialized AI agents for code generation, debugging, planning, and code review using Gemini API.
- Implemented Retrieval-Augmented Generation (RAG) pipeline using Qdrant vector database for contextual code retrieval.
- Developed responsive frontend using React and Next.js for real-time conversational interaction.
- Designed scalable FastAPI backend for agent communication and workflow management.
- Containerized services using Docker and deployed the application on GCP cloud infrastructure.
- Worked on prompt engineering, workflow orchestration, vector embeddings, and conversational AI pipelines.

Technologies Used:

React, Next.js, FastAPI, LangChain, LangGraph, Gemini API, Qdrant, Docker, GCP

GitHub: <https://github.com/yourusername/ai-multiagent-coding-assistant>

Multimodal RAG Chatbot using Gemini LLM

- Developed an end-to-end Multimodal Retrieval-Augmented Generation (RAG) chatbot capable of answering user queries using uploaded documents and contextual retrieval.
- Built an interactive frontend using React for real-time chatbot interaction.

- Implemented semantic vector retrieval using PostgreSQL with pgvector for efficient similarity search and metadata filtering.
- Integrated Gemini LLM for contextual response generation and conversational AI capabilities.
- Designed document embedding and retrieval pipelines for accurate context-aware responses.
- Worked on prompt engineering, chunking strategies, and embedding optimization for improved answer quality.

Technologies Used : React, Python, PostgreSQL, pgvector, Gemini LLM, NLP, Vector Database

GitHub: <https://github.com/yourusername/ai-multiagent-coding-assistant>

Pneumonia Detection using Deep Learning

- Built medical image classification model using CNN and EfficientNet architecture
- Performed image preprocessing and augmentation using OpenCV
- Trained model on chest X-ray dataset for pneumonia detection
- Deployed it in Streamlit

Technologies Used : React, Python, PostgreSQL, pgvector, Gemini LLM, NLP, Vector Database

GitHub: <https://github.com/yourusername/ai-multiagent-coding-assistant>

EXPERIENCE

Data Science Trainee (Self Learning & Projects)

Dec 2022 – Present

- Built and deployed multiple machine learning and deep learning projects
- Gained hands-on experience in model development, evaluation, and deployment
- Worked with real-world datasets for NLP and recommendation systems
- Used Docker and AWS EC2 for application deployment

Data Science Intern – AI Variant

May 2023 – Jan 2024

- Worked on projects including recommendation systems and fake news detection
- Applied supervised and unsupervised learning techniques
- Gained practical exposure to machine learning workflows

CERTIFICATIONS

- Completed Data Science Course at ExcelR - 2023
- Tata Data Visualization: Empowering Business with Effective Insights (Forage) - 2025
- British Airways Data Science Virtual Experience (Forage) – 2025

LANGUAGES

- Tamil - Native
- English - Professional Proficiency